Stochastic Processes
Spring 2025 January 14, 2025

Homework #1: Conditional Expectation
Due: February 13, noon

#1 **(Chapter 3, Exercise 3)** The joint probability mass function of X and Y, $p(x, y)$ is given by

$$\begin{array}{lll} p(1,1) = \frac{1}{9} & p(2,1) = \frac{1}{3} & p(3,1) = \frac{1}{9} \\ p(1,2) = \frac{1}{9} & p(2,2) = 0 & p(3,2) = \frac{1}{18} \\ p(1,3) = 0 & p(2,3) = \frac{1}{6} & p(3,3) = \frac{1}{9} \end{array}$$

Compute $E[X|Y = i]$ for $i = 1, 2, 3$.

#2 **(A Trading Model)** The fundamental value of the security is a random variable V taking two possible states H(high) and L(low) and $P(V = H) = p_0$. There are two types of traders in this market: informed traders and uninformed traders. The proportion of informed traders is $\alpha \in (0, 1)$. Once the market opens, the state of V is realized and informed traders get to know V. They will buy a share of the security if $V = H$ and sell it otherwise. Uninformed traders do not know the state of V. They will buy or sell the security randomly with equal probability. Compute the conditional probability that $V = H$ if you observe that a trader is trying to buy the security.

#3 **(Chapter 3, Exercise 21)**Consider Example 3.12, which refers to a miner trapped in a mine. Let N denote the total number of doors selected before the miner reaches safety. Also, let T_i denote the travel time corresponding to the ith choice, $i \geq 1$. Again let X denote the time when the miner reaches safety.

(a) Give an identity that relates X to N and the T_i.

(b) What is $E[N]$?

(c) What is $E[T_N]$?

(d) What is $E[\sum_{i=1}^{N} T_i | N = n]$?

(e) Using the preceding, what is $E[X]$?

#4 **(Chapter 3, Exercise 26)** You have two opponents with whom you alternate play. Whenever you play A, you win with probability p_A; whenever you play B, you win with probability p_B, where $p_B > p_A$. If your objective is to minimize the expected number of games you need to play to win two in a row, should you start with A or with B?
Hint: Let $E[N_i]$ denote the mean number of games needed if you initially play i. Derive an expression for $E[N_A]$ that involves $E[N_B]$; write down the equivalent expression for $E[N_B]$ and then subtract.

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